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Digital inclusion, transaction costs, and rural revitalization industrialization: Evidence from rural e-commerce clusters in China

Luyan Huang [a], Zhan Cao [b,*], Dongxun Liu [c]

[a] School of Marxism, Zhengzhou Normal University, Zhengzhou, Henan 450000, China
[b] Zhejiang Financial College, Hangzhou 310018, China
[c] Key Research Institute of Yellow River Civilization and Sustainable Development, Henan University, Zhengzhou 450000, China

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ABSTRACT

Based on panel data from 2007 to 2023 on regions in China with rural e-commerce clusters, this paper investigates the relationship among digital inclusion, transaction costs, and the industrialization of rural revitalization. The results indicate that, first, digital inclusion promotes the industrialization of rural revitalization; second, transaction costs exert a significant inhibitory effect on the industrialization of rural revitalization, specifically in that increases in information acquisition costs and transportation costs both impede the industrialization process. Furthermore, by introducing transaction costs as a mediating variable, the study finds that transaction costs partially mediate the impact of digital inclusion on the industrialization of rural revitalization, with both information acquisition costs and transportation costs serving as mediating pathways.

1. Introduction

With the rapid development of the digital economy, the role of digital technologies in promoting urban-rural integration and rural revitalization has become increasingly prominent. Especially in rural areas of China, the rise of e-commerce has provided new momentum for optimizing agricultural industrial structures, expanding marketing channels for agricultural products, and increasing farmers' incomes (Basnayake et al., 2024). Against this backdrop, digital inclusion has become a critical foundation for achieving rural digital transformation and industrial upgrading. Digital inclusion not only refers to the widespread availability of internet infrastructure but also encompasses rural residents' proficiency in digital technologies and their practical application abilities in areas such as e-commerce, payments, and information access. By increasing rural areas' participation in the digital economy, digital inclusion is expected to effectively drive the industrialization process of rural revitalization.

At the same time, rural market development is still limited by transaction costs. In conventional agricultural transactions, problems such as information asymmetry, low logistics efficiency, and slow market response lead to high information acquisition and transportation costs, thereby restricting the marketization of rural industries (Xu et al., 2024). Consequently, examining how digital transformation can help reduce these transaction costs holds important practical significance for advancing rural industrial development.

Although existing literature has addressed the relationship among the digital economy, rural e-commerce, and rural development

* Corresponding author.
E-mail addresses: luie1989@163.com (L. Huang), gipsycao@163.com (Z. Cao), liudx@henu.edu.cn (D. Liu).

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to some extent, research gaps remain, primarily in two areas: first, a lack of in-depth analysis of the mechanisms by which digital inclusion drives rural industrialization; second, the oversight of transaction costs as an intermediary variable in this process. Accordingly, this paper systematically investigates, both theoretically and empirically, how digital inclusion, by influencing transaction costs, affects the level of industrialization in rural revitalization.

To this end, this paper is structured around the following research hypotheses: First, digital inclusion promotes the development of rural e-commerce, thereby advancing rural industrial revitalization; second, increased transaction costs hinder the process of rural industrial revitalization, specifically reflected in information acquisition costs and transportation costs; finally, transaction costs serve as a mediating factor between digital inclusion and rural industrial revitalization, with information acquisition costs and transportation costs each constituting key mediation pathways. Through empirical analysis of rural e-commerce clusters in China, this study seeks to uncover the underlying mechanisms of rural industrial revitalization in the digital context and to provide both theoretical and empirical support for formulating more inclusive and efficient rural development policies.

2. Research questions and hypotheses

Digital inclusion refers to the extent to which residents in rural areas can access and effectively participate in digital economic activities on an equal footing (Ye and Jiang, 2025). With the widespread adoption of internet access and the improvement of digital skills, rural residents are able to participate more actively in e-commerce, online transactions, and the exchange of market information, thus promoting the development of rural e-commerce (Borjigin et al., 2025). This not only provides more market opportunities for rural areas but also facilitates the modernization of agricultural production methods and the extension of industrial chains (Cen et al., 2022). As digital inclusion increases, rural enterprises can leverage e-commerce platforms to expand their markets and increase the added value of agricultural products, thereby driving the process of rural industrial revitalization (Xu et al., 2025). Therefore, digital inclusion enhances the level of rural industrial revitalization by promoting the adoption and participation in e-commerce platforms.

H1: Digital inclusion promotes rural industrial revitalization.

Transaction costs refer to the various expenses and resource consumption incurred to complete transactions in the market process (Tim et al., 2021). High transaction costs—particularly those related to information acquisition and logistics—reduce the market efficiency of rural e-commerce, limiting the expansion opportunities for rural enterprises and the circulation capacity of agricultural products (Wang, 2023). Information asymmetry and high logistics costs often result in inefficient allocation of market resources, thus impeding the process of rural industrialization (Huang et al., 2023). Therefore, it is hypothesized that transaction costs have an inhibitory effect on the industrialization of rural revitalization; an increase in transaction costs will render market operations less efficient and constrain the development of industrialization.

H2: The increase in transaction costs inhibits the process of industrialization in rural revitalization.

Information acquisition cost is an important component of transaction costs (Wei et al., 2025). Excessive information acquisition costs may prevent rural residents and enterprises from obtaining market information—such as product prices, demand changes, and supply chains—in a timely manner, thereby affecting their production decisions and level of market participation (Liu and Li, 2024). In the context of rural e-commerce, high information acquisition costs can prevent rural products from promptly meeting market demand and may even affect price transparency and fair trading, ultimately hampering the process of rural industrialization (Liu et al., 2023). Thus, an increase in information acquisition costs directly impacts market liquidity and transaction efficiency, thereby restricting the advancement of industrialization in rural revitalization.

H2a: An increase in information acquisition costs inhibits the process of industrialization in rural revitalization.

Transportation costs constitute another significant aspect of transaction costs. In rural areas, due to relatively underdeveloped transport infrastructure, logistics costs are typically high (Wen et al., 2024). Elevated transportation costs result in low circulation efficiency for rural products, restrict the market reach of agricultural goods, and impede the process of industrialization (Zhang et al., 2023a). If transportation expenses are excessively high, even high-quality agricultural products from rural areas may struggle to effectively access broader markets, thereby constraining economic diversification and industrial integration in rural economies (Fu and Guo, 2025). Therefore, increased transportation costs inhibit the development of rural industrialization and reduce market competitiveness.

H2b: Rising transportation costs hinder the progress of industrialization in rural revitalization.

Digital inclusion can mitigate transaction costs by enhancing network access rates, the adoption of electronic payments, and digital skills in rural areas, thereby reducing information acquisition costs and improving logistics efficiency (Wang et al., 2023). In an environment of lower transaction costs, rural enterprises can engage in market transactions more efficiently, thus fostering industrialization (Bi, 2024). Consequently, transaction costs play a critical mediating role between digital inclusion and rural revitalization industrialization; digital inclusion promotes the industrialization process of rural revitalization by reducing transaction costs.

H3: Transaction costs serve as a mediating factor between digital inclusion and rural revitalization industrialization.

Information acquisition costs are a core component of transaction costs (Lu et al., 2025). With the advancement of digital inclusion, rural residents and businesses can more readily access market information via the internet and digital platforms, which reduces the degree of information asymmetry and decreases transaction costs (Liu et al., 2024). Improved transparency and timeliness of information allow rural e-commerce to better align with market demand, thereby increasing product added value and market competitiveness (Zhai et al., 2023). Therefore, information acquisition costs play a key mediating role between digital inclusion and rural revitalization industrialization. Digital inclusion facilitates the development and industrialization of rural e-commerce by lowering the cost of information acquisition.

H3a: Information acquisition costs mediate the relationship between digital inclusion and rural revitalization industrialization.

Digital inclusion is reflected not only in information access and electronic payment but may also reduce transportation costs by promoting the development of logistics technologies (Lv et al., 2025). As digital inclusion increases, the integration between logistics systems and e-commerce platforms becomes closer, which helps to reduce transportation costs and improves the circulation efficiency of agricultural products (Peng, 2022). Lower transportation costs can enhance the market accessibility of rural products and promote the diversification and clustering of rural industries (Zhang et al., 2023b). Therefore, transportation costs also mediate the relationship between digital inclusion and rural revitalization industrialization. Digital inclusion, by optimizing logistics and reducing transportation costs, thereby facilitates the process of rural revitalization industrialization.

H3b: Transportation costs mediate the relationship between digital inclusion and rural revitalization industrialization.

3. Theoretical framework of variables

The explained variable is Rural Industrialization for Revitalization (Rural). Rural industrialization for revitalization refers to the advancement of competitive rural industries so as to promote the modernization, diversification, and overall upgrading of agriculture and the rural economy. Its primary objective is to accelerate the transformation of agricultural production models, enhance the added value of agricultural products, increase farmers' incomes, and improve the quality of rural life through industrialization. In this paper, the degree of rural industrialization for revitalization is measured by the proportion of rural enterprises engaged in agriculture, processing, manufacturing, retail, and other industries relative to the total number of rural enterprises.

The core explanatory variable is Digital Inclusion (Digital_Inclusion).which reflects the extent to which different social groups, particularly rural residents, can access and participate fairly in digital economic activities. It includes internet connectivity, digital literacy, and engagement in e-payment and e-commerce. To capture this concept, the paper measures digital inclusion by the number of network connections per household in rural e-commerce clusters across China, as this indicator effectively reflects both accessibility and the level of digital participation.

In this study, transaction costs (Transaction_costs) are treated as a mediating variable, referring to the expenses and resources consumed in market exchanges, mainly information acquisition costs and transportation costs. Information costs (inform_costs) are measured by the logarithm of market information expenses in rural e-commerce clusters, while transportation costs (trans_costs) are proxied by regional transport infrastructure, where better infrastructure implies lower costs. This approach is adopted because these two dimensions represent the most critical barriers affecting rural market efficiency and industrial development.

To accurately examine the effects of digital inclusion and transaction costs on rural industrialization, this paper introduces several control variables. Regional economic development (GDP) is considered, as it shapes industrial progress. Human capital (human_capital) and average wage (wage) reflect digital skills and participation capacity, while government support (gov) significantly influences rural revitalization. Population density (des) is linked to market size (market) and transaction costs, facilitating industrial agglomeration in densely populated areas. Infrastructure construction (infra), particularly transport and logistics, directly affects industrial efficiency, and agricultural technology (tech) enhances productivity and product quality. Controlling these variables ensures robust, accurate empirical analysis.

4. Research methods and data description

4.1. Research methods

This paper constructs a baseline regression model, as shown in Eqs. (1) and (2):

$$\text{Rural}_{i,t} = \alpha_0 + \alpha_1 \text{Digital_Inclusion}_{i,t} + \alpha_n \textit{Controls}_{i,t} + \partial_i + \theta_t + \varepsilon_{i,t} \tag{1}$$

$$\text{Rural}_{i,t} = \rho_0 + \rho_1 \text{Transaction_costs}_{i,t} + \rho_n \textit{Controls}_{i,t} + \partial_i + \theta_t + \varepsilon_{i,t} \tag{2}$$

This paper constructs a mediation effect regression model, as shown in Eqs. (3) and (4):

$$\text{Transaction_costs}_{i,t} = \gamma_0 + \gamma_1 \text{Digital_Inclusion}_{i,t} + \gamma_n \textit{Controls}_{i,t} + \partial_i + \theta_t + \varepsilon_{i,t} \tag{3}$$

Table 1
Results of Data Statistical Analysis.

VarName	Obs	Mean	SD	Min	Median	Max
Rural	40,034	0.3206	0.1213	0.0541	0.3207	0.8223
Digital_inclusion	40,034	0.6720	0.5311	0.0029	0.5449	4.8941
inform_costs	40,034	0.7719	0.4583	0.0008	0.6768	5.6911
trans_costs	40,034	0.7591	0.2785	0.0233	0.7902	1.8318
infra	40,034	13.8512	1.2781	2.6391	13.9566	19.4429
tech	40,034	10.2594	1.0754	0.0000	10.3500	12.7875
gdp	40,034	10.4674	0.8730	7.6324	10.4687	17.5154
human_capital	40,034	0.0202	0.0248	0.0000	0.0109	0.1848
wage	40,034	10.7837	0.7366	6.7719	10.7387	15.5240
gov	40,034	0.2199	0.2135	0.0032	0.1633	3.8290
des	40,034	0.0691	0.2269	0.0000	0.0287	4.7260
market	40,034	22.1877	1.4374	13.8738	22.2633	26.7317

Table 2
Baseline regression results of digital inclusion and rural revitalization industrialization.

VARIABLES	(1) Rural	(2)	(3)	(4)	(5)
Digital_inclusion	0.0020***	0.0026***	0.0020***	0.0020***	0.0021***
	(2.8990)	(3.8339)	(3.0574)	(3.0224)	(3.1621)
infra		−0.0057***	−0.0059***	−0.0059***	−0.0061***
		(−16.5728)	(−17.2843)	(−17.1610)	(−18.4757)
tech		−0.0042***	−0.0044***	−0.0044***	−0.0047***
		(−4.3422)	(−4.5124)	(−4.4666)	(−4.8512)
gdp			0.0028***	0.0027***	0.0011
			(4.1195)	(3.9330)	(1.5758)
human_capital			0.1661***	0.1652***	0.1594***
			(3.6538)	(3.6398)	(3.5922)
wage				0.0008	0.0002
				(1.0346)	(0.2916)
gov				−0.0008	0.0022
				(−0.5077)	(1.4039)
des					−0.0197*
					(−1.7261)
market					0.0063***
					(6.8509)
Constant	0.3192***	0.4416***	0.4138***	0.4057***	0.2968***
	(663.0425)	(43.3110)	(36.7398)	(29.6411)	(14.1737)
Obs	40,034	40,034	40,034	40,034	40,034
County	YES				
Year	YES				
R-squared	0.9523	0.9524	0.9535	0.9536	0.9535

$$\text{Rural}_{i,t} = \vartheta_0 + \vartheta_1 \text{Digital_Inclusion}_{i,t} + \vartheta_2 \text{Transaction_costs}_{i,t} + \vartheta_n \textit{Controls}_{i,t} + \partial_i + \theta_t + \varepsilon_{i,t} \tag{4}$$

4.2. Data statistical analysis

The descriptive statistics in Table 1 show that the dataset is complete and reliable. The average level of rural revitalization industrialization (Rural) is 0.3206, ranging from 0.0541 to 0.8223, revealing clear regional disparities. Digital inclusion (Digital_inclusion) has a mean of 0.6720, with wide variation across regions (maximum 4.8941), indicating uneven digital access and usage. The mediating variables—information acquisition cost (0.7719) and transportation cost (0.7591)—are relatively high, suggesting notable transaction barriers in rural areas. Control variables, including infrastructure (infra), technology (tech), GDP, human capital, wages, and government support, show reasonable distributions, ensuring robust data for subsequent regression analysis.

5. Empirical results analysis

5.1. Baseline regression analysis

5.1.1. Baseline regression analysis of digital inclusion

The baseline regression results in Table 2 demonstrate that the core explanatory variable—digital inclusion (Digital_Inclusion)—has a clear and significant positive impact on the industrialization of rural revitalization (Rural) across all model specifications (e.g., 0.0020 in column (1) and 0.0021 in column (5)). This indicates that higher levels of internet penetration and improved digital

Table 3
Baseline analysis results of information costs and rural revitalization industrialization.

VARIABLES	(1) Rural	(2) Rural	(3) Rural	(4) Rural	(5) Rural
inform_costs	−0.0047***	−0.0038***	−0.0032***	−0.0028**	−0.0032***
	(−4.0686)	(−3.4174)	(−2.7491)	(−2.4669)	(−2.8130)
infra		−0.0058***	−0.0057***	−0.0054***	−0.0061***
		(−15.2806)	(−14.2615)	(−13.5966)	(−15.9628)
tech		−0.0104***	−0.0085***	−0.0085***	−0.0085***
		(−10.0052)	(−7.3394)	(−7.2220)	(−7.4058)
gdp			0.0052***	0.0050***	0.0016**
			(6.7284)	(6.3734)	(2.0551)
human_capital			0.3304***	0.3272***	0.2859***
			(7.1165)	(7.0701)	(6.3459)
wage				0.0040***	0.0029***
				(4.4784)	(3.6061)
gov				0.0005	0.0065***
				(0.2744)	(3.6845)
des					−0.0118
					(−1.5023)
market					0.0121***
					(12.9152)
Constant	0.3095***	0.4937***	0.4273***	0.3816***	0.1717***
	(331.0749)	(45.3148)	(32.3761)	(22.7208)	(7.2616)
Obs	40,034	40,034	40,034	40,034	40,034
County	YES	YES	YES	YES	YES
Year	YES	YES	YES	YES	YES
R-squared	0.9473	0.9484	0.9434	0.9435	0.9446

Table 4
Baseline regression results of transportation costs and rural revitalization industrialization.

VARIABLES	(1) Rural	(2) Rural	(3) Rural	(4) Rural	(5) Rural
trans_costs	−0.0305***	−0.0383***	−0.0338***	−0.0349***	−0.0338***
	(−5.7457)	(−7.0750)	(−6.5270)	(−6.6891)	(−6.6071)
infra		−0.0053***	−0.0058***	−0.0055***	−0.0062***
		(−13.0689)	(−14.5557)	(−13.8332)	(−16.1221)
tech		−0.0090***	−0.0091***	−0.0092***	−0.0092***
		(−7.5832)	(−7.8796)	(−7.7858)	(−8.0029)
gdp			0.0053***	0.0051***	0.0018**
			(6.9188)	(6.6044)	(2.3294)
human_capital			0.2893***	0.2835***	0.2474***
			(6.4255)	(6.3323)	(5.6317)
wage				0.0045***	0.0034***
				(4.9411)	(4.1274)
gov				0.0015	0.0074***
				(0.7928)	(4.1037)
des					−0.0126
					(−1.6090)
market					0.0119***
					(13.0071)
Constant	0.2975***	0.4575***	0.4076***	0.3557***	0.1501***
	(73.7230)	(36.3362)	(30.1953)	(20.5362)	(6.2067)
Obs	40,034	40,034	40,034	40,034	40,034
County	YES	YES	YES	YES	YES
Year	YES	YES	YES	YES	YES
R-squared	0.9423	0.9434	0.9435	0.9436	0.9447

inclusion in rural areas contribute to optimizing the industrial structure and markedly foster the progress of rural industrialization. Hence, the empirical evidence strongly validates Hypothesis H1, which posits that digital inclusion facilitates rural e-commerce development and thereby advances rural industrial revitalization.

5.1.2. Baseline regression analysis of transaction costs

According to the regression results presented in Table 3, information acquisition costs (inform_costs) consistently exhibit a significant negative effect on rural revitalization industrialization (Rural) across all model specifications, remaining robust at the 1 % significance level. For example, in column (1), the coefficient for information acquisition costs is −0.0047, while in column (5) it is significantly −0.0032. This indicates that as the cost of obtaining market information increases, the efficiency of rural e-commerce and

Table 5
Results of the endogeneity analysis.

VARIABLES	(1) Phase one Digital_inclusion	(2) Phase two Rural	(3) Phase one inform_costs	(4) Phase two Rural	(5) Phase one trans_costs	(6) Phase two Rural
L.Digital_inclusion	0.8635*** (72.0797)					
Digital_inclusion		0.0807*** (45.8127)				
L.inform_costs			1.0107*** (14.0933)			
inform_costs				−0.0028** (−1.9940)		
L.trans_costs					0.9805*** (84.5279)	
trans_costs						−0.0339*** (−9.9845)
infra	0.0112*** (8.6617)	−0.0123*** (−17.1505)	0.0009* (1.7180)	−0.0048*** (−6.7062)	0.0009*** (3.5993)	−0.0062*** (−8.5995)
tech	−0.0084*** (−8.2822)	0.0028*** (4.7614)	0.0014*** (3.4965)	−0.0007 (−1.1162)	0.0004* (1.8993)	−0.0016*** (−2.6436)
gdp	0.0159*** (9.5960)	−0.0033*** (−3.4978)	0.0013* (1.8688)	0.0081*** (8.2582)	−0.0025*** (−7.6781)	0.0108*** (11.7124)
human_capital	1.0974*** (23.9260)	−0.5076*** (−19.6384)	−0.1083*** (−6.3997)	−0.0268 (−1.1049)	−0.0048 (−0.5878)	−0.0317 (−1.3109)
wage	−0.0007 (−0.3781)	0.0090*** (9.2732)	−0.0036*** (−4.6443)	0.0240*** (25.5543)	−0.0013*** (−3.5280)	0.0247*** (26.4108)
gov	0.0109* (1.7915)	−0.0029 (−0.8965)	−0.0055** (−2.2373)	0.0273*** (8.3641)	−0.0027** (−2.2523)	0.0302*** (9.2777)
des	0.0213*** (4.5224)	0.0168*** (6.2825)	−0.0045** (−2.3609)	0.0213*** (7.7788)	0.0013 (1.3754)	0.0190*** (6.9492)
market	0.0018 (1.5231)	0.0283*** (41.9259)	0.0005 (1.0759)	0.0311*** (44.6693)	0.0011*** (4.6685)	0.0290*** (41.3810)
Constant	3.4985*** (4.0959)	−0.2705*** (−17.5145)	3.4902*** (4.8959)	−0.6478*** (−48.3499)	0.0958*** (10.9804)	−0.6322*** (−47.1776)
LM statistic	2.7e+04		3.6e+04		3.7e+04	
Wald F statistic	1.0e+05		1.0e+06		1.6e+06	
Obs	37,595	37,595	37,595	37,595	37,595	37,595
County	YES	YES	YES	YES	YES	YES
Year	YES	YES	YES	YES	YES	YES
R-squared	0.7743	0.2153	0.9703	0.1753	0.9833	0.1804

the development of related industries are inhibited, thereby constraining the advancement of rural revitalization industrialization. Thus, Hypothesis H2a, "An increase in information acquisition costs inhibits the progress of rural revitalization industrialization," receives significant empirical support.

According to the regression results presented in Table 4, transportation costs (trans_costs) consistently exhibit a significant negative impact on rural revitalization industrialization (Rural) across all columns, remaining robust at the 1 % significance level. For instance, the coefficient is significantly −0.0305 in column (1), and remains −0.0338 in column (5). This indicates that as transportation costs rise in rural areas, logistics efficiency declines, the circulation of agricultural products is impeded, and the expansion and upgrading of rural industries are constrained, thereby inhibiting the process of rural revitalization industrialization. These findings provide strong support for Hypothesis H2b: Increasing transportation costs inhibit the process of rural revitalization industrialization.

Combined with the analysis results in Table 4 regarding information acquisition costs, information acquisition costs (inform_costs) likewise demonstrate a stable and significant negative effect on rural revitalization industrialization at the 1 % significance level. Therefore, Hypothesis H2 can be further validated: In general, an increase in transaction costs suppresses the process of rural revitalization industrialization, with the main mechanisms reflected specifically in the dimensions of information acquisition and transportation.

5.2. Endogeneity analysis

To address potential endogeneity issues with the core variables, this study employs the instrumental variable approach and conducts two-stage least squares (2SLS) regression analysis. The specific results are presented in Table 5. For digital inclusion, information acquisition cost, and transportation cost, their respective first-order lagged terms (L.Digital_inclusion, L.inform_costs, L.trans_costs) are introduced as instrumental variables. In the first-stage regression, the instrumental variables all have a significant impact on the corresponding endogenous variables, indicating strong relevance and meeting the correlation requirement for instrumental variables. In the second-stage regression, the core explanatory variables continue to exert significant influence on rural revitalization industrialization, both in terms of direction and magnitude (for example, the coefficient of Digital_inclusion on Rural is 0.0807, significant at

Table 6
Results of mechanism analysis.

VARIABLES	(1) inform_costs	(2) Rural	(3) trans_costs	(4) Rural
Digital_inclusion	−0.0407***	0.0020***	−0.0128***	0.0024***
	(−10.2347)	(3.0187)	(−11.0754)	(3.6411)
inform_costs		−0.0024**		
		(−2.4113)		
trans_costs				−0.0260***
				(−5.9547)
infra	−0.0072***	−0.0061***	0.0020***	−0.0062***
	(−3.9139)	(−18.4039)	(3.7777)	(−18.5874)
tech	0.0100***	−0.0048***	0.0153***	−0.0051***
	(2.8545)	(−4.8813)	(10.4711)	(−5.2717)
gdp	0.0317***	0.0010	−0.0060***	0.0013*
	(7.0960)	(1.4644)	(−5.2976)	(1.8168)
human_capital	−2.4643***	0.1654***	1.3843***	0.1234***
	(−10.6777)	(3.7398)	(15.5689)	(2.8590)
wage	−0.0411***	0.0003	−0.0132***	0.0005
	(−10.0965)	(0.4386)	(−8.8944)	(0.7897)
gov	−0.0922***	0.0024	−0.0109***	0.0025
	(−10.3770)	(1.5504)	(−4.4601)	(1.5877)
des	−0.0942***	−0.0194*	0.0234***	−0.0203*
	(−4.4558)	(−1.6997)	(3.2136)	(−1.7843)
market	0.0110**	0.0063***	0.0034***	0.0062***
	(2.4124)	(6.7942)	(2.7633)	(6.8816)
Constant	0.6845***	0.2951***	0.6862***	0.2789***
	(6.6433)	(14.1344)	(22.1247)	(13.0205)
Obs	40,034	40,034	40,034	40,034
County	YES	YES	YES	YES
Year	YES	YES	YES	YES
R-squared	0.8733	0.9534	0.9655	0.9536

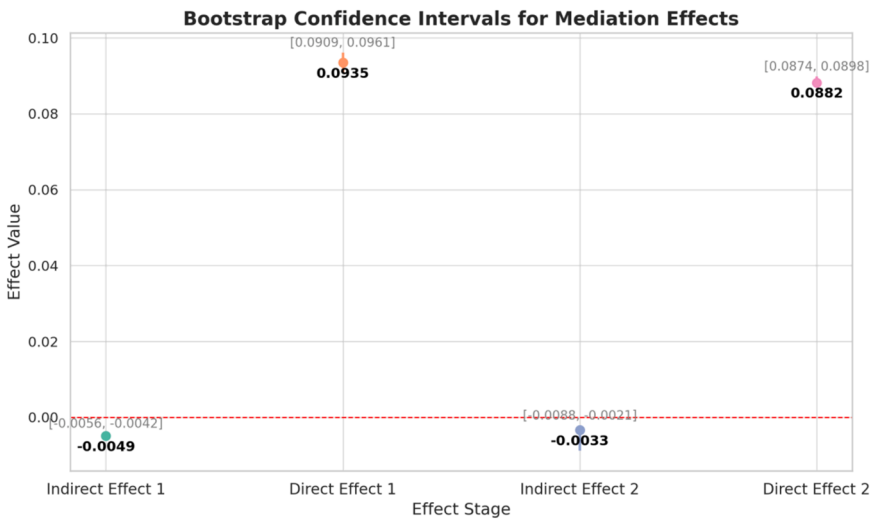

Fig. 1. Bootstrap test chart.

the 1 % level). Moreover, both the LM statistic and the Wald F test support the validity of the model. Therefore, the key findings of this study are not subject to serious endogeneity bias, and the results are robust.

5.3. Mechanism analysis

According to the results of the mechanism analysis presented in Table 6, digital inclusion (Digital_inclusion) exerts an indirect impact on the industrialization of rural revitalization by influencing transaction costs. Column (1) indicates that digital inclusion significantly reduces information acquisition costs (the coefficient of Digital_inclusion is −0.0407); column (2) further demonstrates that information acquisition costs have a significant negative effect on the level of industrialization (the coefficient of inform_costs is −0.0024). At the same time, digital inclusion maintains a significantly positive relationship with industrialization, thereby confirming

H3a: information acquisition costs play a mediating role between digital inclusion and the industrialization of rural revitalization. Similarly, column (3) shows that digital inclusion significantly reduces transportation costs (coefficient: −0.0128), and column (4) indicates that transportation costs have a significant negative effect on industrialization (coefficient: −0.0260), while the positive significance of digital inclusion remains, thereby validating H3b: transportation costs also have a mediating effect between the two. Digital inclusion can indirectly promote the industrialization of rural revitalization by reducing information and transportation costs within transaction costs. This suggests that the improvement of digital infrastructure and the proliferation of e-commerce platforms are conducive to alleviating information asymmetry and logistics barriers in rural markets, thereby enhancing the efficiency of rural industries and their access to markets. Hypothesis H3 is therefore fully supported.

Fig. 1 presents the results of the mediating effect test for digital inclusion impacting rural revitalization industrialization through transaction costs using the Bootstrap method. The four effect phases in the figure represent the indirect and direct effects corresponding to two mediating pathways (information acquisition cost and transportation cost). As shown in the figure, the 95 % confidence intervals for Indirect Effect 1 (−0.0049) and Indirect Effect 2 (−0.0033) are both entirely below zero, indicating that both mediating pathways exhibit significant negative mediating effects. In contrast, Direct Effect 1 (0.0935) and Direct Effect 2 (0.0882) are both positive, and their confidence intervals are entirely above zero, demonstrating that after controlling for transaction costs, digital inclusion still has a significant positive impact on the industrialization of rural revitalization. In summary, Fig. 1 supports the existence of a "partial mediation" mechanism, thus verifying the hypotheses H3, H3a, and H3b proposed in this paper.

6. Conclusions

Based on panel data from China's rural e-commerce clusters (2007–2023), this study systematically explores the relationship between digital inclusion, transaction costs, and the industrialization of rural revitalization. The findings show that digital inclusion significantly promotes rural industrialization by enhancing internet access, digital skills, and participation in e-commerce, while high transaction costs—especially information acquisition and transportation costs—significantly inhibit industrial development by reducing efficiency and limiting market reach. Further analysis reveals that transaction costs play a partial mediating role, as digital inclusion indirectly drives industrial upgrading by lowering information and logistics expenses. Therefore, the government should accelerate rural digital infrastructure construction, enhance network access and digital services, strengthen farmers' digital skills, improve logistics systems, establish low-cost information service platforms, and promote the integration of rural e-commerce with agriculture, processing, and tourism to advance rural revitalization.

CRediT authorship contribution statement

Luyan Huang: Writing – original draft, Supervision, Resources, Methodology, Data curation. **Zhan Cao:** Writing – original draft, Funding acquisition, Formal analysis. **Dongxun Liu:** Writing – review & editing, Conceptualization.

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Data availability

The authors do not have permission to share data.

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